

ZERO

ONE

Welcome

ZERO
ONE
HACK
AI HACKATHON

**ZERO
ONE
HACK**

MAY 29 - 31
2026



VIENNA,
AUSTRIA





Zero One Hack

ZERO
ONE
HACK
AI HACKATHON

36 hours.

Real model training.

Not prompt engineering.

120 curated builders. European supercompute.

Fine-tune and deploy real models on 64 × A100s, not wrappers.



• **May 29–31**

Vienna, Austria

• **36h**

Build · Train · Pitch

• **120**

Selected builders

• **€10K+**

Prize pool · 3 tracks

Partners

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A big thank you to our *partners*





AI Factory Austria

AI:AT

24.4.2026



Europe's Challenge is not AI Capability – It is **Integrating** and **Adoption at Scale**

01 Adoption & Execution

- Lack of structured adoption pathways
- Limited ability to realize beyond pilots

02 Translation Gap

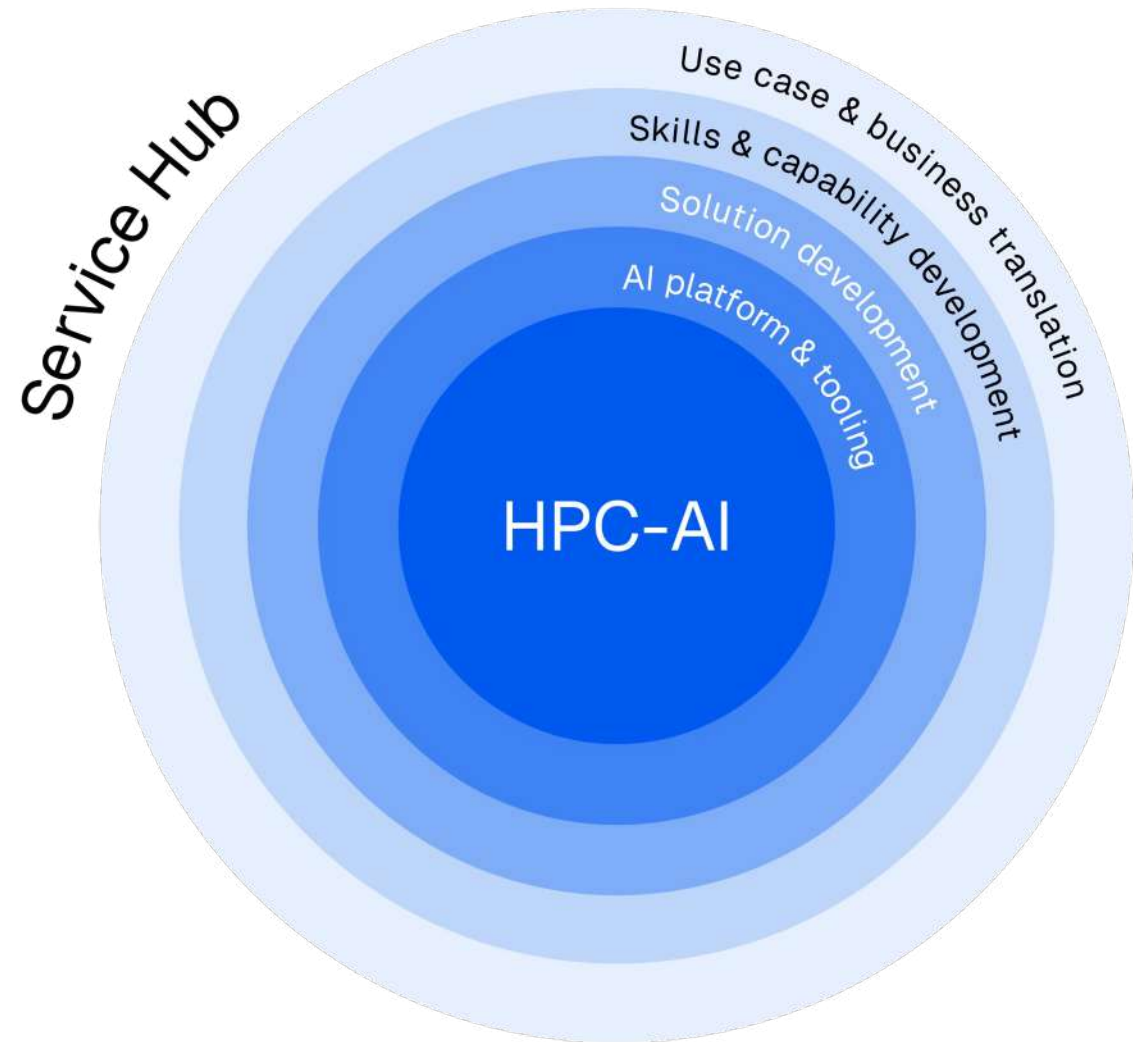
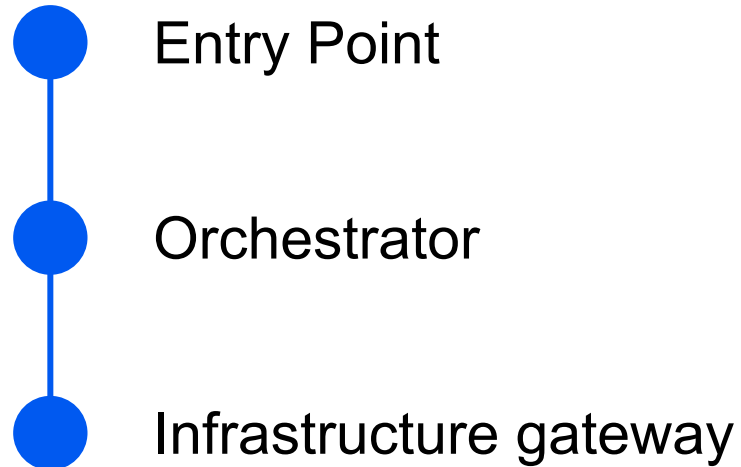
- Gap between research and real-world impact

03 Access & Infrastructure

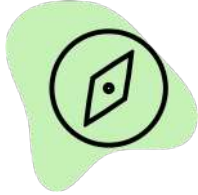
- Fragmented access to AI infrastructure
- Increasing need for sovereign AI capabilities

AI Factories are designed to close this gap by combining infrastructure, hands-on expertise, and orientation

From AI adoption to **sovereign infrastructure** — an integrated service mode



Need for Orientation



Mainly SMEs, Public Administration

- Neutral initial orientation and advice
- Training and Identification of relevant use cases
- Involvement of suitable ecosystem partners
- Support for the first AI pilot projects

Need for Sovereignty



Mainly Public Administration, Critical Infrastructure, Industry

- Access to sovereign infrastructure
- Trusted AI Environments
- Expertise from research, industry and ecosystem

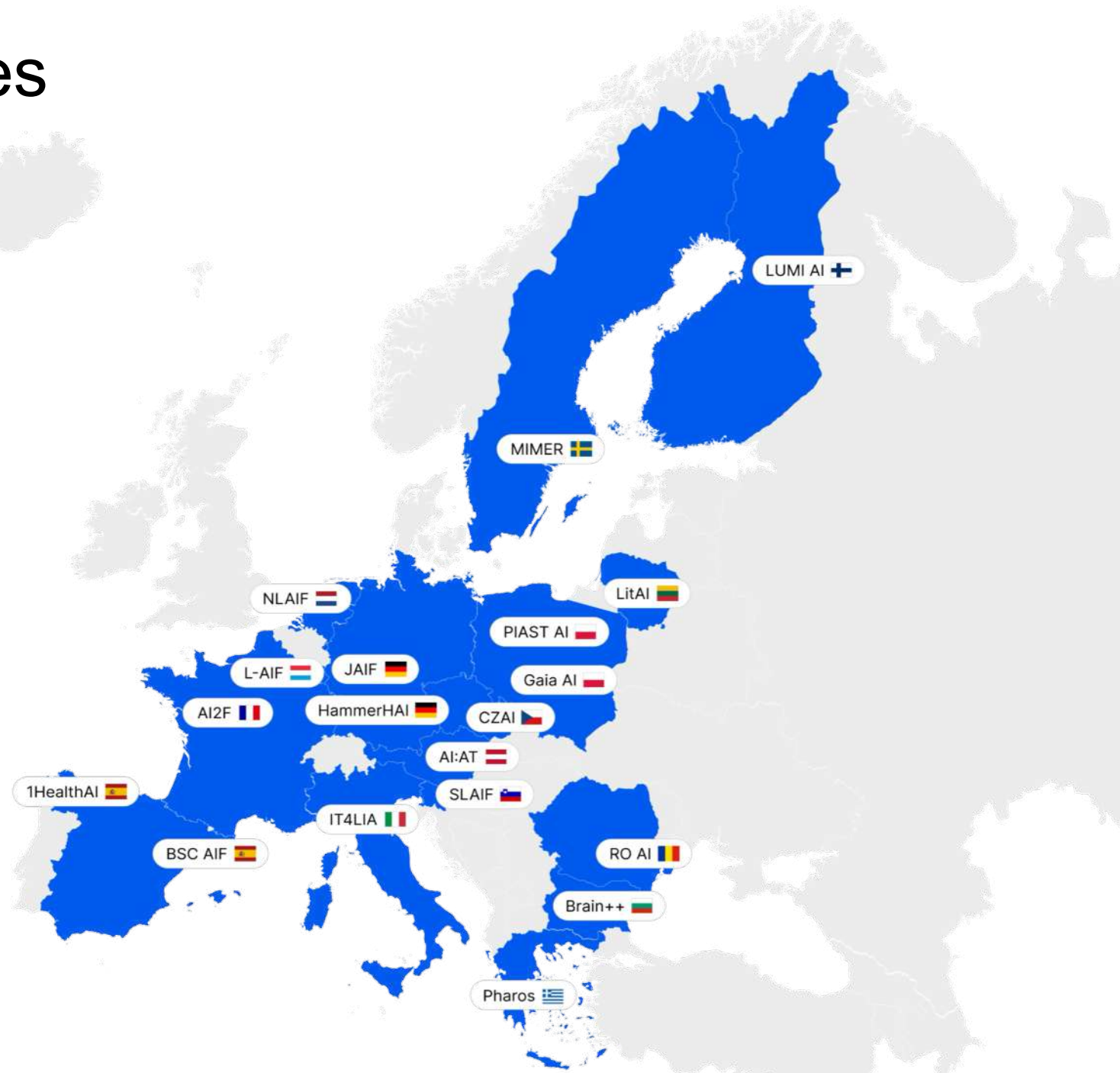
High AI Maturity



Mainly Start-Ups, Deep Tech, Spin-Offs, Industry, Applied Research

- Access to HPC infrastructure
- Expertise in AI Engineering & Trustworthy AI
- Coworking
- Support for access to funding

19 AI Factories



GPU Hardware – EuroHPC Systems



AI:AT Consortium

Beneficiaries



Affiliated Entities



universität
wien



Funded by



EuroHPC
Joint Undertaking



**Co-funded by
the European Union**

 **Federal Ministry
Innovation, Mobility
and Infrastructure
Republic of Austria**

under discussion with



AI Factory Austria AI:AT has received funding from the European High-Performance Computing Joint Undertaking (JU) under grant agreement No 101253078. The JU receives support from the Horizon Europe Programm of the European Union and Austria (BMIMI / FFG).

Contact

Simeon Harrison

Tech Lead

Team AI & Business Growth
AI Factory Austria AI:AT

info@ai-at.eu

Karl-Farkass-Gasse 22, 1030 Wien



@ai-factory-austria



HP E



KI verändert unsere Welt.
Jetzt ist deine Chance, sie mitzugestalten.

42 AUSTRIA

**BEYOND
THE
CoDE**

Abwarten ist keine Strategie - Abwarten bedeutet Rückstand.
Vorausdenken bedeutet jetzt investieren -
die Dividenden sind die IT-Talente der Zukunft.

42 Global

BEYOND
THE
CoDE



42 Global

WHY 42

Digitalisation = Industrial Revolution 2.0

Need for future ICT skills in Austria to compete in a extremely fast-changing global economy

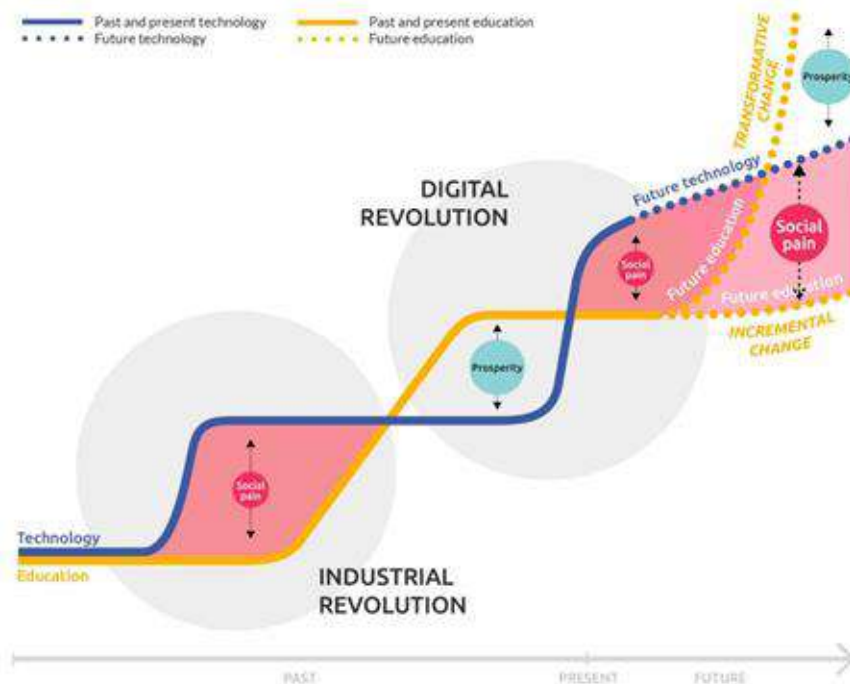
AI as a disruptor and reshapes business models, products and the way we work and live

Talent shortage (AT & DE) - Austria already faces a shortage of around 28,000 ICT specialists, in Germany, this gap could reach ~660,000 by 2040 (sources WKO, bitkom)

→ 42 offers a radically different, scalable way to develop these future-ready digital talents.



OECD LEARNING COMPASS 2030



Abwarten ist keine Strategie - Abwarten bedeutet Rückstand.
Vorausdenken bedeutet jetzt investieren -
die Dividenden sind die IT-Talente der Zukunft.

42 Global

A GLOBAL NETWORK



50+

Campuses

28

Average age

30+

Countries

20%

Women

61k

Trained

2013

Founded

27k

Active

96%

Work in the IT field

Digitale Kompetenzen sind Infrastruktur. 42 aktiviert verborgenes Potenzial und transformiert es in produktive Skills.

42 Learning Modell

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THE
CoDE**



42 Learning Modell

THE CURRICULUM

COMMON CORE | INNER CIRCLE

Level 1-9

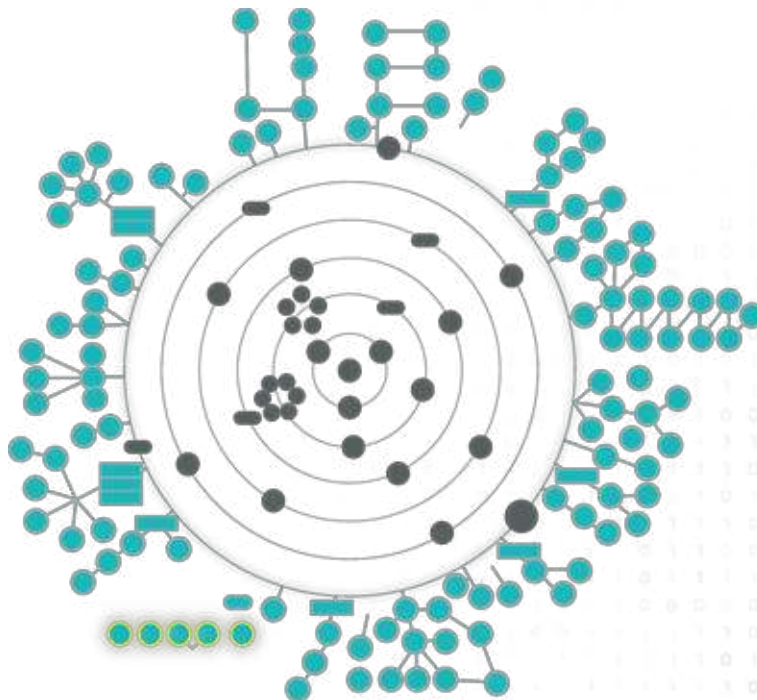
Solid foundational knowledge:

- C (imperative programming)
- Python (object-oriented programming)
- AI-Projects
- System administration & networks
- Web development

16 units (individual and team projects) of increasing project complexity

5 exams

Regular updates
by 42 Central



ADVANCED | OUTER CIRCLE

Level 10-21

Selection of 2-3 specialisations, e.g.:

- Algorithms, AI & Data Science
- Security & Encryption
- Gaming, Creatives, Graphics and UX
- System Admin and Network
- Kernel and OS
- Web & Mobile App Development

Individual, team & group projects, boot camps, etc.

CAMPUS-SPECIFIC SPECIALISATIONS

Agile Software Dev.(RBI), ISTQBready (RBI), SAP Students Zone

42 Learning Modell

THE STATISTIC

640

Students total

296

Active students

112

Common Core
validations

21.8 %

of women

29

Years old
on average

50.5 %

dropped out

15

piscines

1.415

pisciners

7

kickoffs

52

work
experiences I&I

89.2 %

CSAT score

94.4 %

Recommendation
rate

42 Learning Modell

THE USP

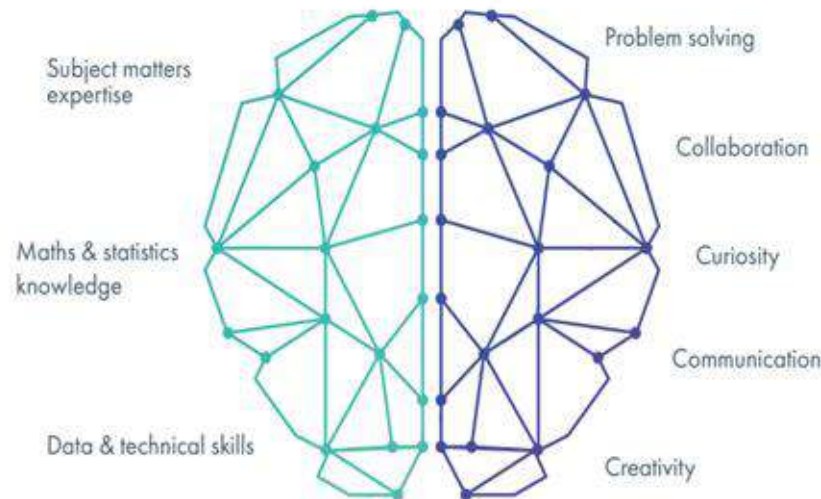
TECH SKILLS

Deep dive into IT, software, algorithms and coding expertise through programming down to kernel, driver, script, container and OS levels in structured languages such as C, C++, Python and JavaScript.



PERSONAL SKILLS

Ability to learn and adapt to rapidly and often unexpectedly developing technologies, as well as independent and critical thinking to find high-quality software solutions



42 Learning Modell

THE DEGREE

Graduates can obtain the RNCP certificate via 42 Paris:

IT Solutions Designer and Developer

(EQF-Level 6 / Bachelor)

- Applicative Development
- Web and Mobile Development

IT Architecture Expert

(EQF-Level 7 / Master)

- Data and Database Architecture
- Network and Information Systems



KI verändert Wertschöpfung spürbar.
Entscheidend sind die Skills der Umsetzer:innen.

42 in Austria

**BEYOND
THE
CoDE**



42 in Austria

STRONG PARTNERS & INVESTORS FOR THE DIGITAL FUTURE

Business Partners



Supporting Partners & Networks



Sponsors



If 39% of today's skills will disappear by 2030*, the Austrian economy needs talent that builds the future — not manages the past.

*World Economic Forum — The Future of Jobs 2025

That's exactly what 42 is there for.

**BEYOND
THE
CoDE**

Florian Brunner

florian@42vienna.com

42 Vienna

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www.42vienna.com

42 Wels

Bauernstraße 11a | 4600 Wels

www.42wels.at





lumos

Vienna's leading student
org in tech

30 + CS / DS / Business
Students

Data Science & AI

since 2019

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Track Partners

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HACK
AI HACKATHON



#INDU
TRIAL



#FORE
CASTING



#INSU
R
ANCE

Track Partners

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AI HACKATHON



#INDU
TRIAL

VIENNA · 29-31 MAY 2025



Shaping an innovative and sustainable future

René Fabien de Montmorency



Infineon at a glance

Growth areas



Energy
green and efficient

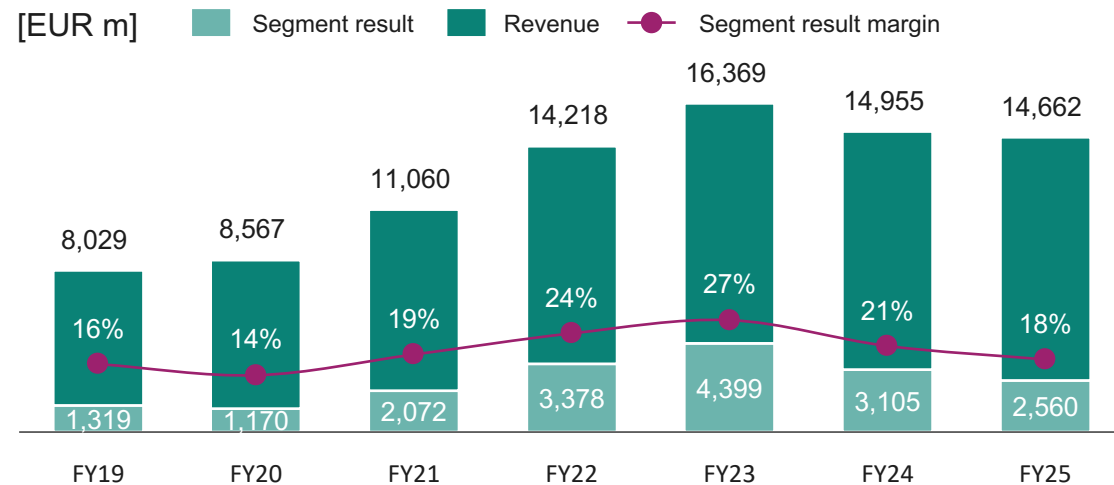


Mobility
clean and safe



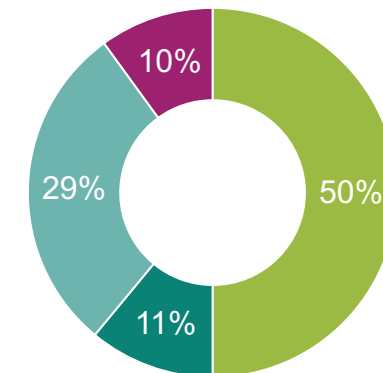
IoT
smart and secure

Financials



FY25 revenue by segment¹

- Automotive (ATV)
- Green Industrial Power (GIP)
- Power & Sensor Systems (PSS)
- Connected Secure Systems (CSS)

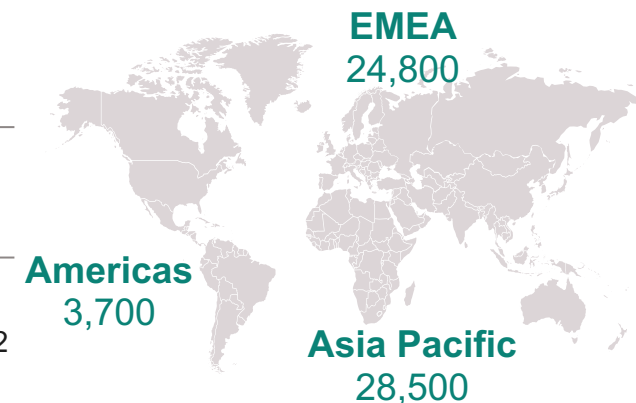


Employees¹

57,000
employees worldwide

75
R&D and

14
manufacturing locations²



For further information: [Infineon Annual Report](#).

¹ 2025 Fiscal year (as of 30 September 2025) | ² As of 30 September 2025

Infineon Austria – Company Overview

Fiscal year 2025 (as of 30 Sep. 2025)

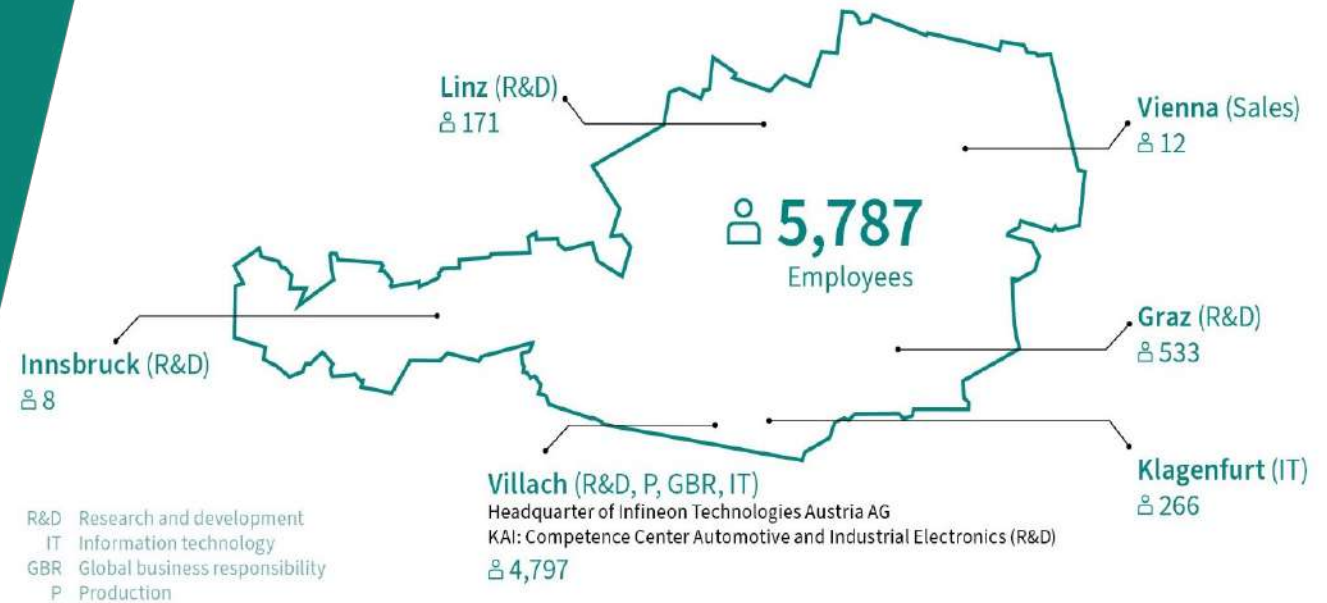
Turnover	€ 4.695	bn
Earnings before taxes	€ - 48	m
Investments	€ 245	m
R&D expenses	€ 721	m

Subsidiaries in Austria

IT Services, Klagenfurt
KAI, Villach

Foreign subsidiaries

Infineon Technologies Romania SCS (R&D)
Infineon Technologies (Kulim) Sdn Bhd, Malaysia (P)
NoBug Consulting SRL, Romania (R&D)
NoBug DOO, Serbia (R&D)

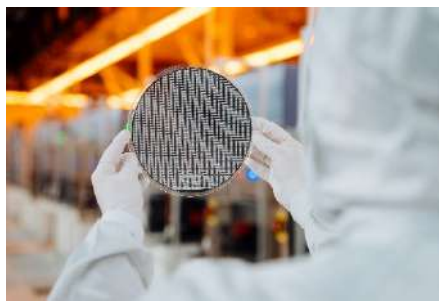


The Innovation Fab Villach for POWER semiconductors

Our 'hat trick'



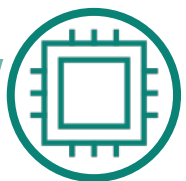
World's thinnest 300 mm silicon power wafer at 20 μm



World's most competitive 200 mm silicon carbide semiconductor factory



World's first 300 mm gallium nitride wafer for power electronics: First products for customers in 10/2025



8 bn chips (GJ 25)



~1,600 Product types



~1.000 processing steps per wafer

Efficient chips for green energy:

Infineon products enable higher efficiency in power flows



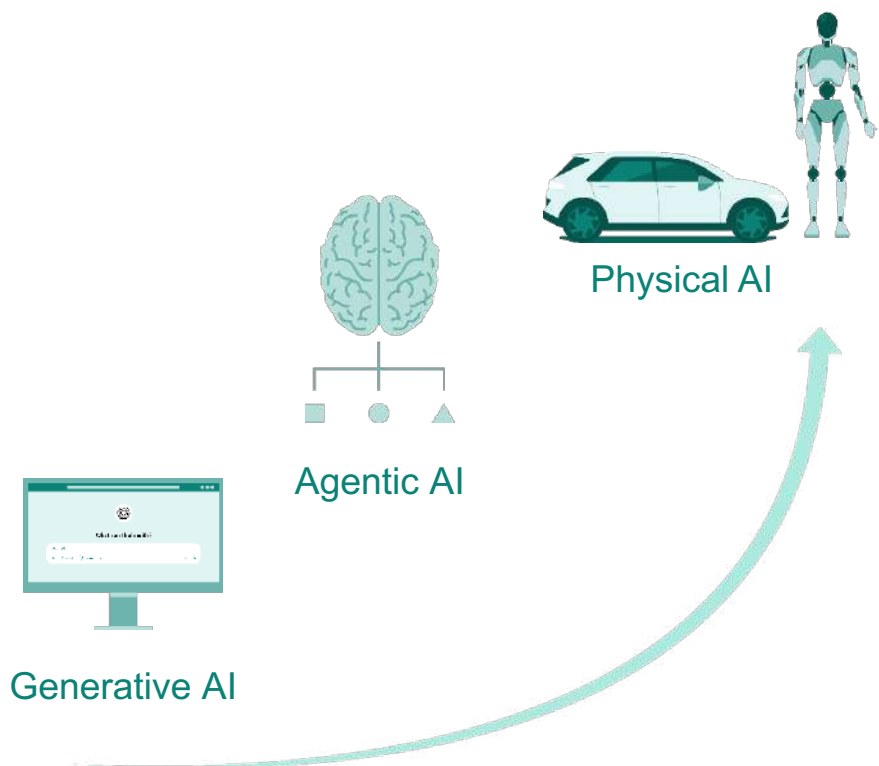
**5% efficiency gain through
Infineon products in a 1 GW
data center**

**Energy for 125,000
households**

1-gigawatt data center; 50 MW efficiency improvement (3,500 kWh per household)

AI as the next technological revolution

Advanced semiconductor solutions are driving this revolution forward.



~730 million

People worldwide will be using AI tools in 2030.

92%

of Fortune 500 companies are using OpenAI technology in 2024.

Artificial intelligence is here to stay.



We power AI



We are creating a more sustainable future by providing technologies that reduce power losses and cooling costs in greener data centers.



We enable and provide AI



We support our customers' innovations with semiconductor solutions, software, and tools that help implement AI innovations quickly, efficiently, and at scale.

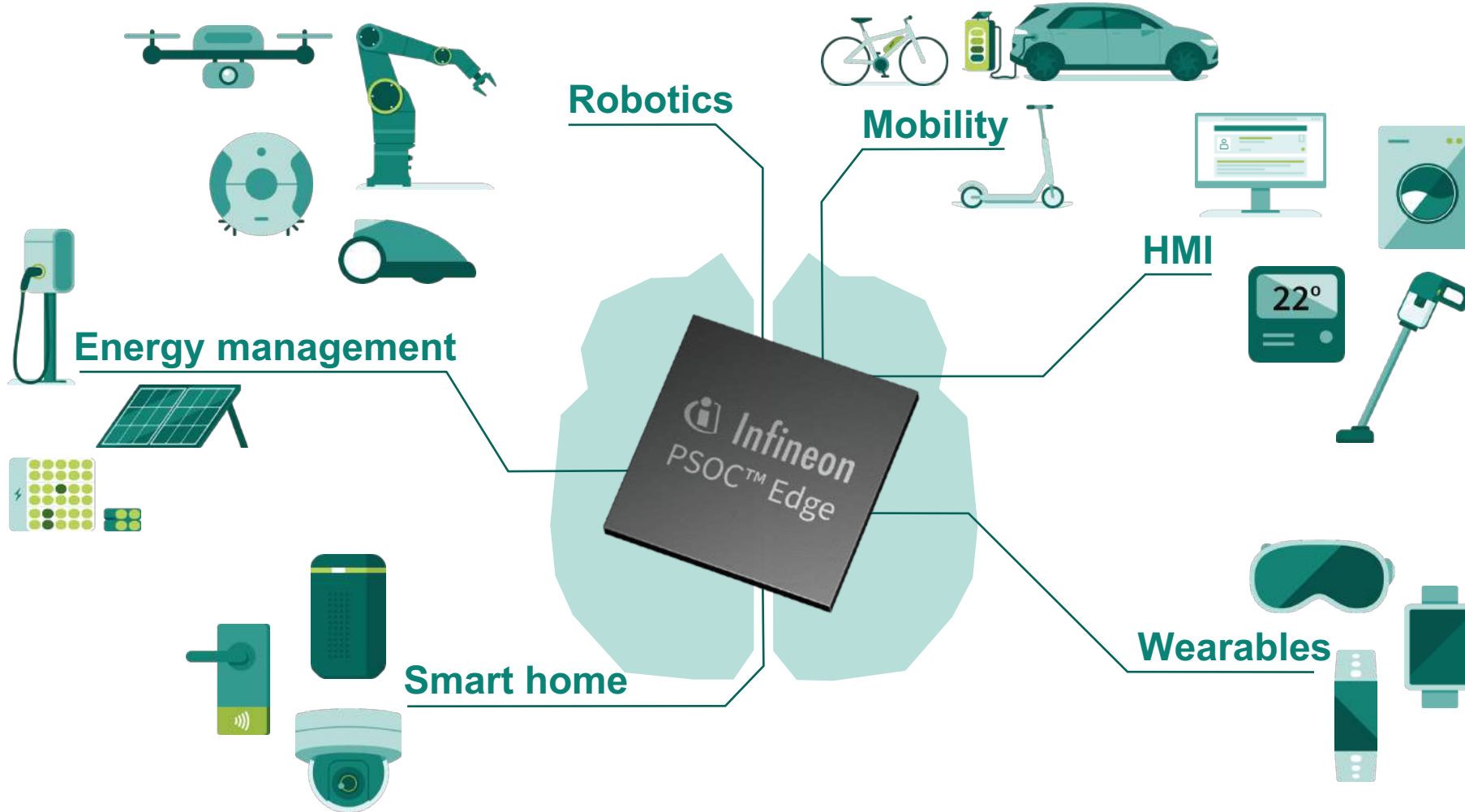


We use AI



We advance high-quality solutions by leveraging AI company-wide to develop smarter products and optimize processes.

PSOC™ Edge



Mobility

- E-Bikes
- Intelligent Charging

Human Machine Interface (HMI)

- Appliances
- Industrial Device Usability
- Factory Automation

Wearables

- Fitness Watch
- AR/MR/VR Glasses & Accessories
- Audio Accessories

Smart Home

- Thermostat
- Speaker
- Door lock
- IP Camera
- Doorbell

Energy management

- Solar panels

Robotics

- Vacuum Cleaner
- Vacuum Robots
- Service Robot
- Lawn & Garden Robotics
- Industrial Robotics

PSOC™ Edge a global story



Powered by Infineon Austria

SECORA™ Pay Green

- R&D competence from Graz
- world's first contactless payment card technology, allowing for up to 100% reduction in plastic waste
- SECORA™ Pay Green cards: eco-friendly Coil-on-Module (eCoM) package – no antenna in the card body needed
- Conventional payment cards: copper wire antenna, that spans the entire card - almost impossible to recycle
- allows the use of recyclable materials



Powered by Infineon Austria

Security chips and Ultra-wideband (UWB) technology

- R&D competence from Graz and Villach

Security chips:

- Applications in contactless payment cards, smart wearables, ID cards and electronic passports

UWB chips:

- UWB technology provides precise and secure localization, sensing and data exchange
- Applications in hands-free devices, "keyless go" systems and "find my" services



Powered by Infineon Austria

Market leader in the relevant radar sectors

- R&D competence from Linz
- **Applications in the automotive sector:**
driver assistance systems such as adaptive cruise control, pedestrian detection, warning systems, automatic emergency brake assistants or blind spot detection
- **Applications in industry, IoT and the consumer market:** interior sensors, smart home systems or drones
- Over 700 million radar chips sold (24/60/77GHz)





Powered by Infineon Austria

Shaping E-Mobility with Leading High Voltage Solutions

- Gallium nitride (GaN), Silicon (Si) and Silicon Carbide (SiC) technology with R&D expertise, and production in Villach
- Supporting higher driving range and affordability at the same time
- Highly efficient charging
- Great driver experience

Powered by Infineon Austria

Quantum Research made in Austria

- Unique quantum test laboratory in Villach
- Ion trap technology for scalable quantum processors
- A reliable partner for research, industry, and companies driving a strong quantum ecosystem, including Quantinuum, IonQ, and others
- Post-quantum cryptography and the world's first Common Criteria certification for a security controller



Partnerships strengthen R&D excellence

Endowed professorships & Academic cooperations

University Zagreb: Cooperation with Faculty of Electrical Engineering and Computing in the field of power electronics

Politecnico Milano: RF Design & Mixed Signal IP

University of Modena: Power Electronic Simulations and Defect Optimizations on GaN

University of Udine: Power Conversion & Wireless Power Transfer

Infineon Quantum Test Lab in Villach

Development of ion-based quantum processors with integrated optical interfaces in cooperation with JOANNEUM Research and the University of Innsbruck



193

Research collaborations in Austria, Europe and beyond (FY 2025)

Strengthening Europe's knowledge-base with IPCEI

Participation in the "Important Project of Common European Interest (IPCEI) on Microelectronics and Communication Technologies (ME/CT)": Industrial Implementation of New Semiconductor Generations and Knowledge Transfer to EU-13 and Balkan Countries

Participation in Silicon Austria Labs

Contribution to a newly developed centre of gravity in electronics-based systems

Selected collaborations



Educational and research networks



Clusters and European projects



Listen2Future



Corporate Social Responsibility



Research Centers



Partners

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EWOR



Featherless 

Red Bull

Ströck

Almdudler

Communication

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Documentation



Discord



Emergency Numbers

Lumos +43 660 3482771

AI:AT +43 676 5118499

For specific contacts, please
refer to the Support section in
the documentation

Need Help?

We've Got You Covered!



Check the Q&A

Your first stop for
general questions.



Lumos Team

Talk to us for quick
fixes.



Open an Issue

Hit a roadblock? Report
it, we'll fix it ASAP

Schedule

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Friday, May 29

18:00-19:00 Registration
19:30-20:30 Kickoff
20:30-22:00 Case Reveal
22:00 Hack Begins

Saturday, May 30

08:00-09:00 Breakfast
11:00-11:45 Panel Discussion
12:00-14:00 Lunch
13:00-17:00 Mentoring Sessions
14:00-14:30 HPE Presentation
19:00-20:00 Dinner

Sunday, May 31

10:00 Submission
10:00-11:00 Breakfast
13:15-13:30 Final Speech
13:30-14:30 Final Pitches
14:45-15:00 Final Results
Until 16:30 Venue Closing

Judges 1/3

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Rene Fabien
De Montmorency



*Infineon: Start-Up and
Partner Management*

Catarina
Bogensperger-Held



*Uniq: Head of Product
Marketing & Research*

Marcel
Moosbrugger



Kugelblitz: Co-Founder

Alexander
Spreckelsen



*Hack Nation: Founding
Head of Venture*

Judges 2/3

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Simeon Harrison



*AI:AT: AI Expert and
Startup Coach*

Bernhard Götzendorfer



AI:AT: Venture Builder

Max Seifert



Sybilion: Chief of Staff

Alexander Maitz



*Hack Nation: Founding
Head of Venture*

Judges 3/3

Philipp Omenitsch



sequel: CTO

Miguel Peixoto



*Sybilion: Founding
Senior Machine
Learning Engineer*

Mentors 1/3

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AI HACKATHON

Alexander Sing



*Data Science Consultant
@ Accenture*

Johannes Oster



*AI Engineering
@ Innovatic Group*

Bernhard Götzendorfer



AI:AT: Venture Builder

Mentors 2/3

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AI HACKATHON

Alexander Maitz



Newsroom: Co-Founder

Marcel
Moosbrugger



Kugelblitz: Co-Founder

Alexander
Spreckelsen



*Hack Nation: Founding
Head of Venture*

Mentors 3/3

ZERO
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HACK
AI HACKATHON

Miguel Peixoto



*Sybilion: Founding
Senior Machine
Learning Engineer*

Panel Discussion

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AI HACKATHON

Simeon Harrison



Moderator

Peter Tutsch



ENTROVY Law: Founder

Iveta Lohovska



*HPE: CTO AI &
HPC*

Jose Antonio
Alatorre



*Main
Sequence:
Founder & CEO*

Saturday, May 30
11:00 – 11:45

11

11

VIENNA · 29-31 MAY 2025

26

HPE HPC Talk

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AI HACKATHON

How does HPC work?

Iveta Lohovska



Saturday, May 30
14:00-14:30

HPE: CTO AI & HPC

Wifi



- Most received personal credentials
- If you didn't you can find them via this QR Code → please mark used ones

Food

Fri, May 29

21:00 Dinner
Wraps (here)

Sat, May 30

9:30 Breakfast
Ströck (here)

12:00 / 13:00 Lunch
Neu Mee

19:30 Dinner
Italian (here)

Sat, May 31

10:00 Breakfast
Ströck (here)

15:00 Lunch
Pizza (here)

Rules

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Respect people. Respect the venue. Hack responsibly.



Don't burn anything



No alcohol



Don't break anything



Be nice to another

Rules

Respect people. Respect the venue. Hack responsibly.



Wear your tags



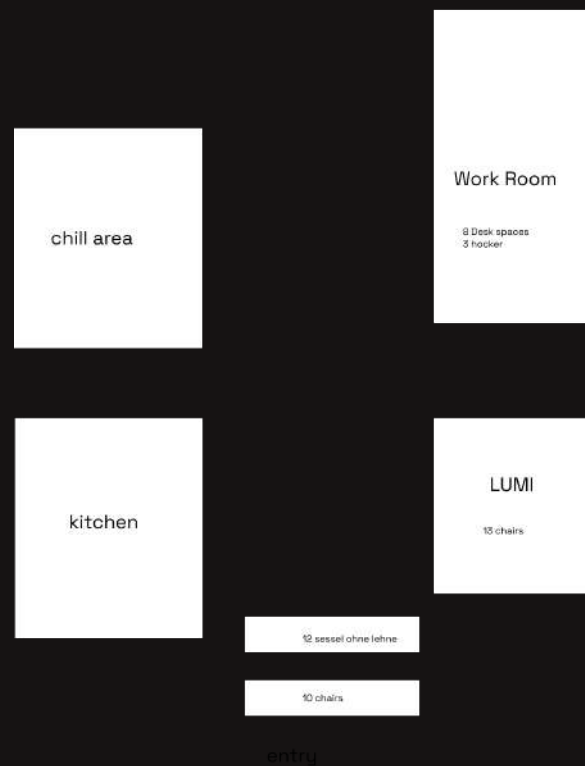
Don't steal

Sleeping/Venue



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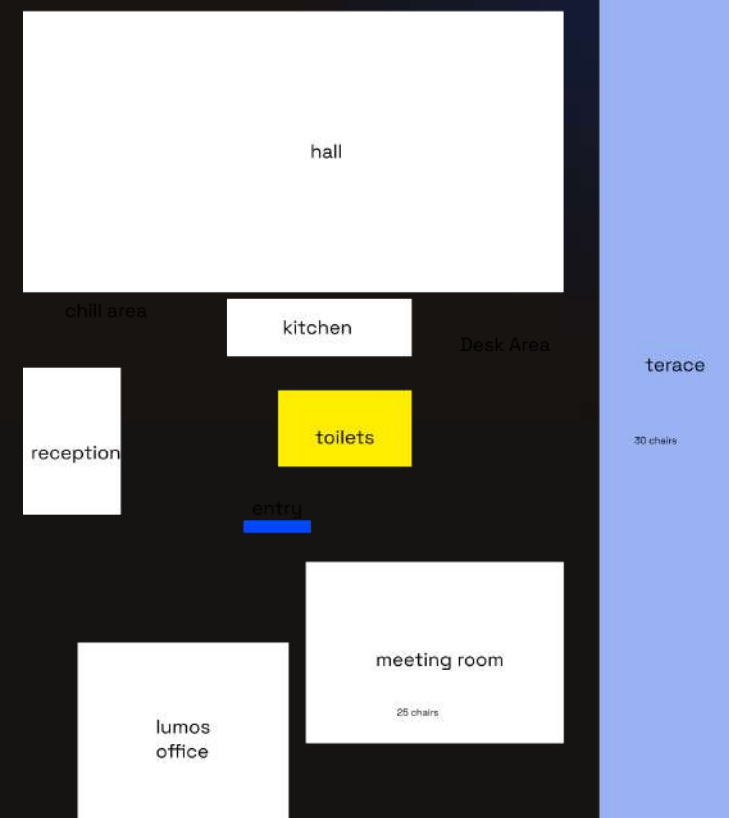
2. Floor



6. Floor



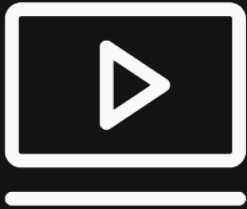
7. Floor



Project Submission

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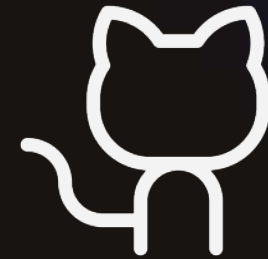
The Three things you need to submit



Slide Deck
(for 3 min)



Demo video
(2 min)



Github Repo
(public)

Project Submission

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By Sunday 10:00, every team submits their work through a single Tally form.

1

Tally Form

2

Technical Review

3

Best Three Teams
Compete for Podium

MIT License

Open source · MIT License · free to use, modify, fork & ship

You keep ownership.

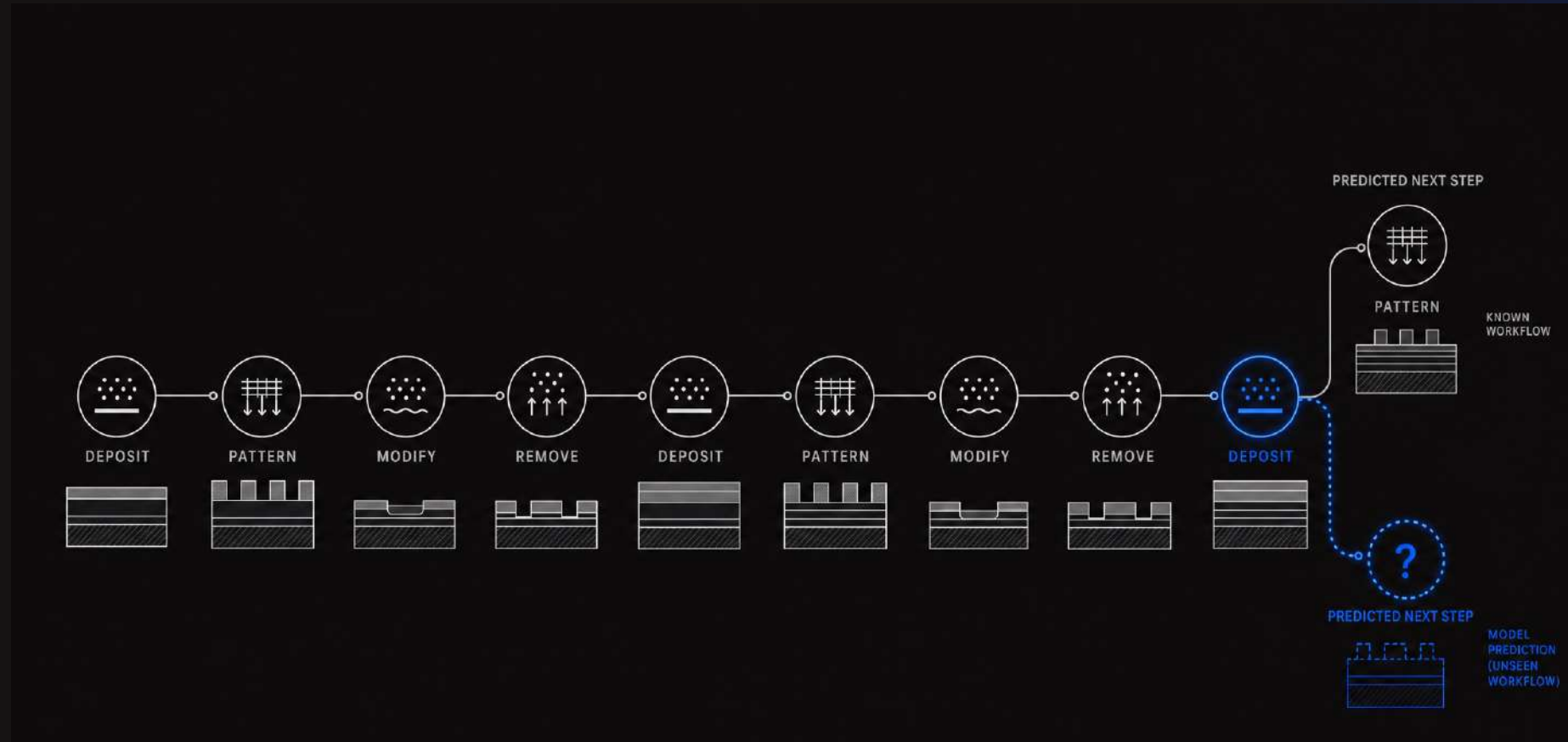
But everyone gets an MIT license to it.

You also keep full rights to your own work.

ZERO ONE

The Tracks

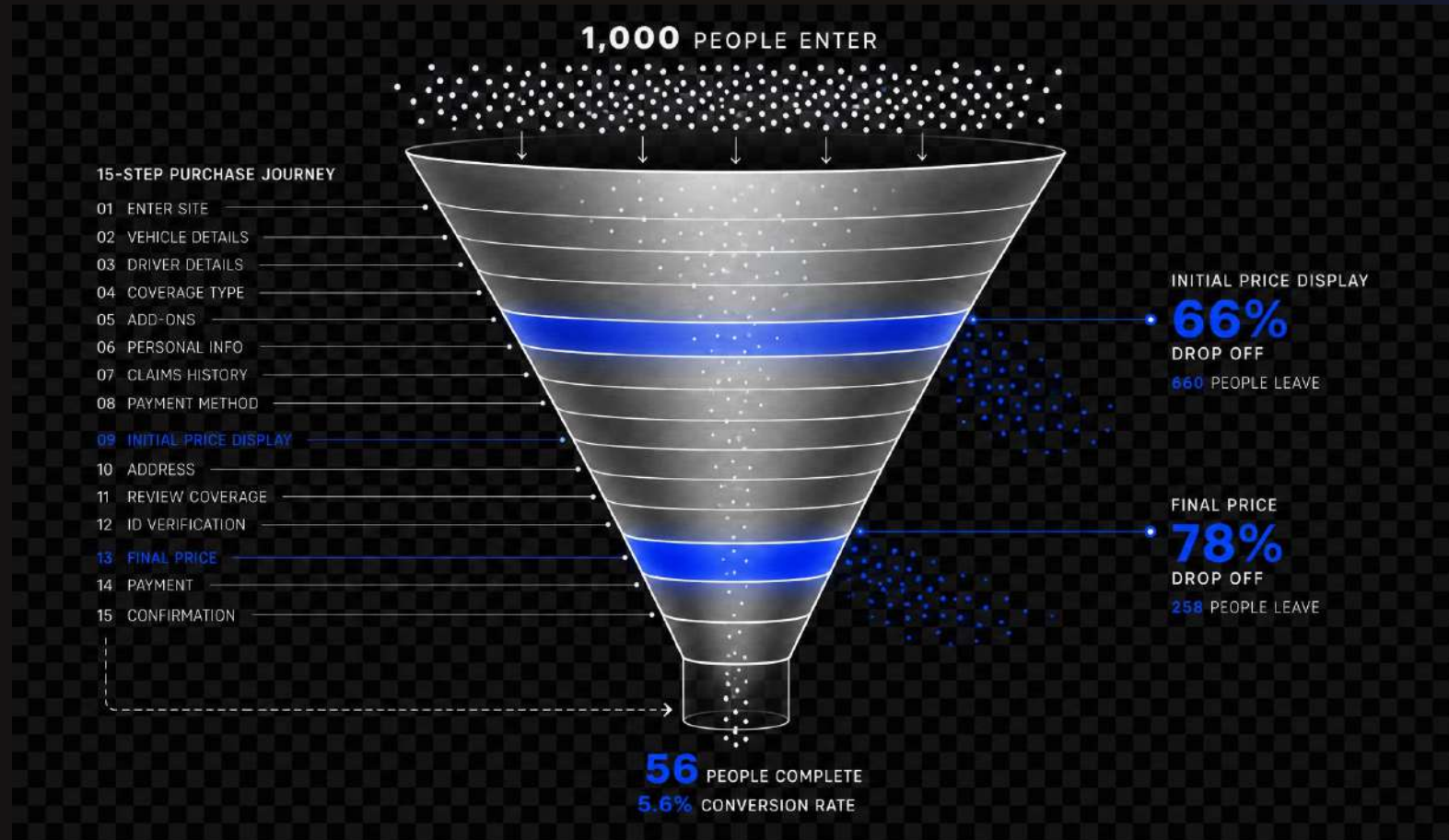
Problem: Real chip fabrication is a 100+ step sequence where order is everything.



Insurance



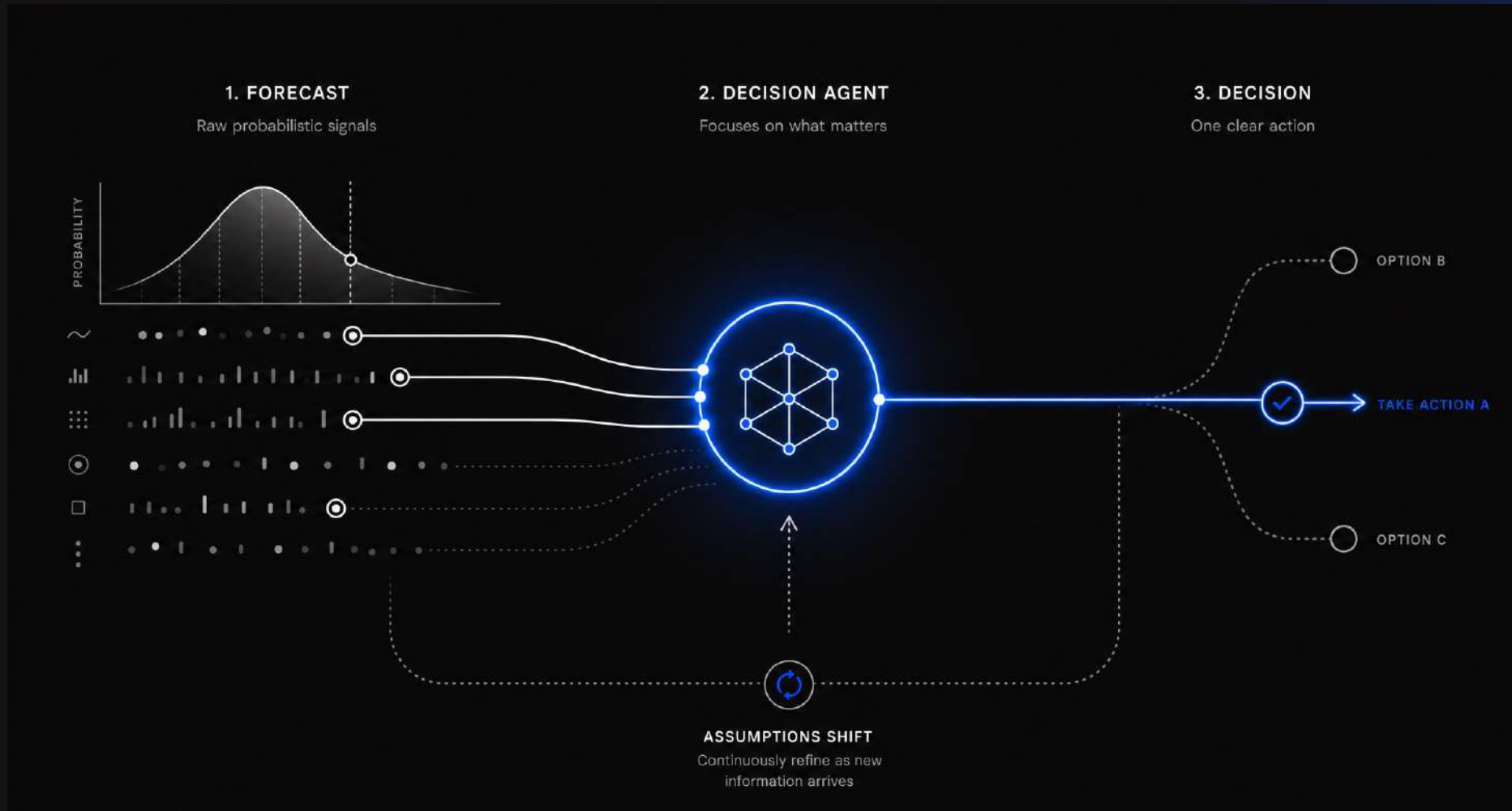
Problem: From 1.000 people who start the online health insurance journey, only 56 finish.



 Sybillion

AI HACKATHON

Problem: Forecasting APIs give you probabilities. Decisions still need a human



How you get access



Via E-Mail / Discord

- > Wi-Fi
- > Featherless



In Person

- > Lovable
- > Leonardo Cluster
- > Sybillion API

What happens next

21:00	Dinner (7th Floor)
21:15	Final Allocation (Teams & Cases
21:00 - 21:30	Computing Workshop's
21:30 -22:00	Tooling Access Handout
22:00	Detailed Case Distribution

**22:00 Hackathon
starts!**

ONBOARDING

HPC

Leonardo

Insurance



For one of the winners Uniqqa will give out one technical Internship

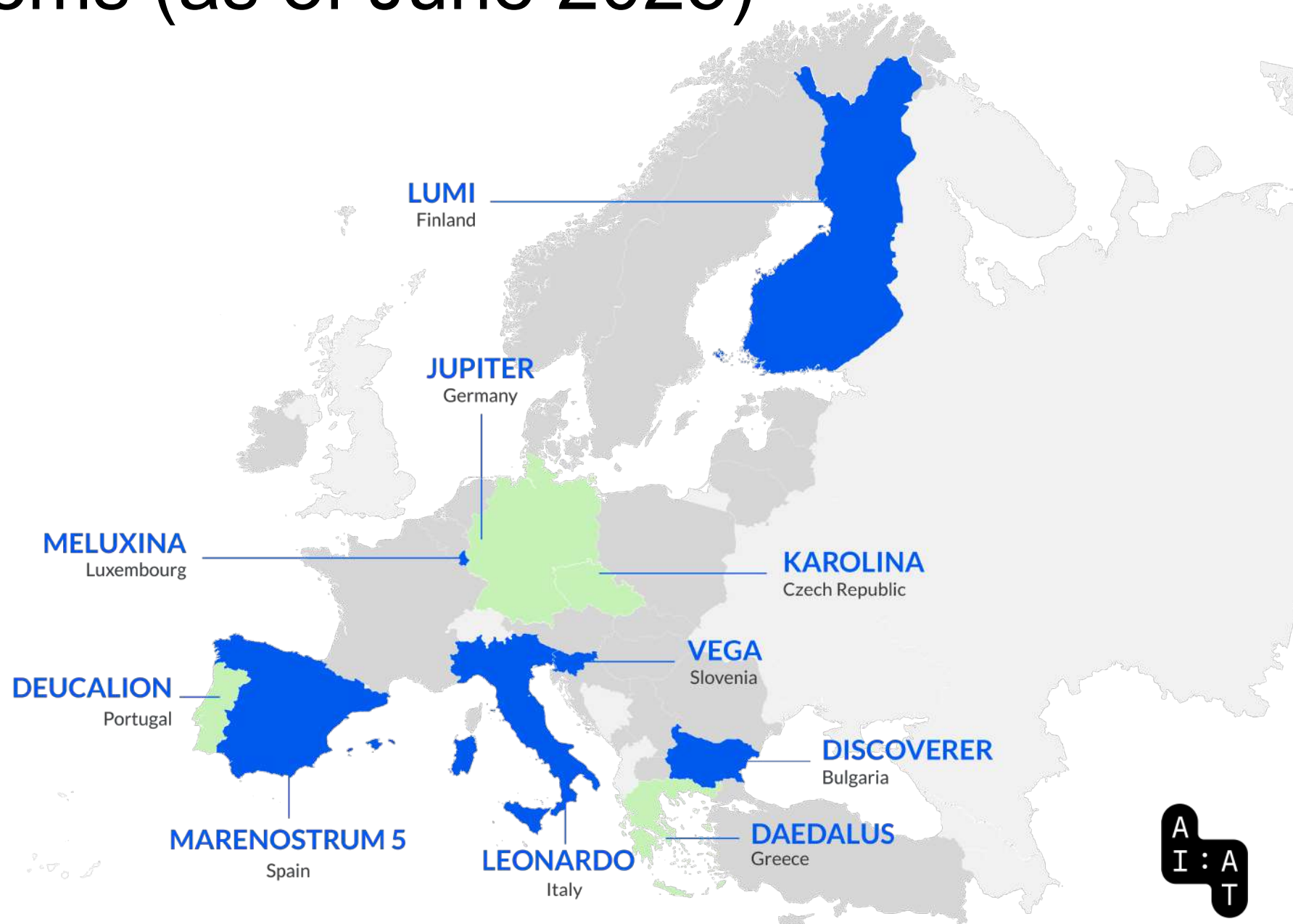
Supercomputer Access at the Hackathon

EuroHPC Systems (as of June 2025)

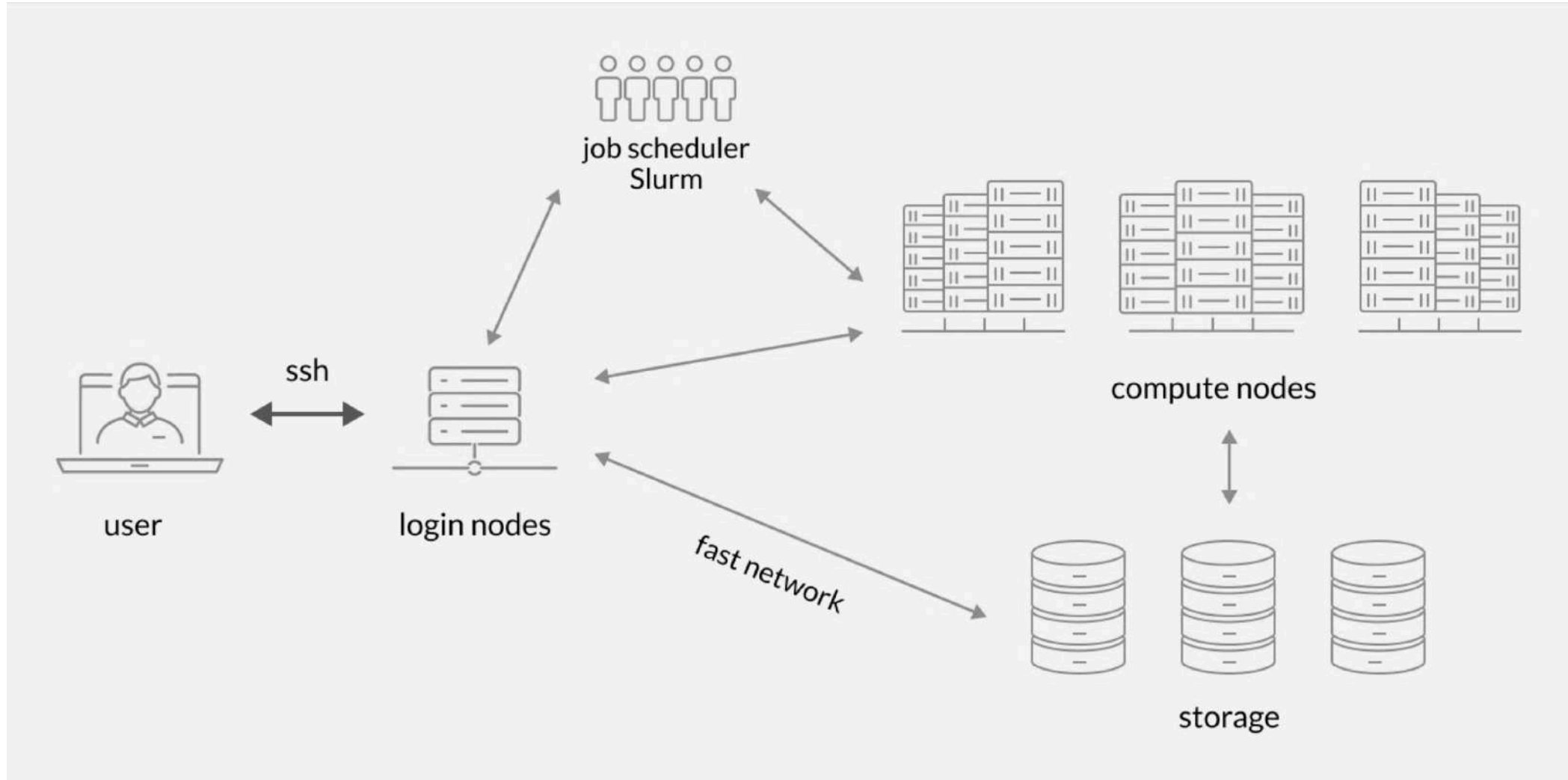
In Top 500 list

<https://www.top500.org/>

- #4 JUPITER
Nvidia GH200
- #9 LUMI
AMD MI250X
- #10 LEONARDO
Nvidia A100



Typical Setup of a Supercomputer



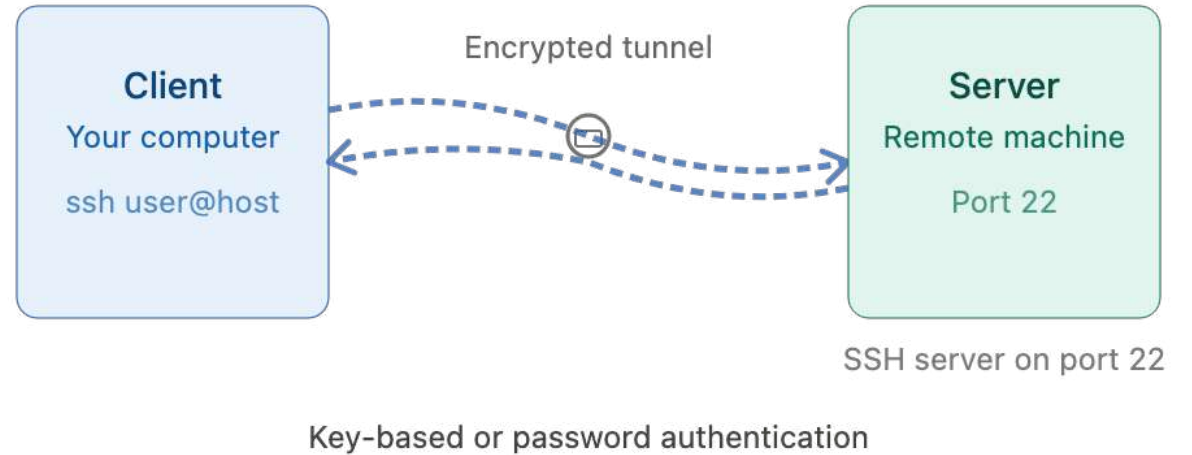
SSH: Secure Shell

What is it?

- Remote access protocol
- End-to-end encrypted

What can you do with it?

- Run commands / get a shell
- File transfer (SCP/SFTP)
- Tunnels / port forwarding



SSH Access to Leonardo

Use any of the following login nodes:

```
ssh your_username@login01-ext.leonardo.cineca.it
```

```
ssh your_username@login02-ext.leonardo.cineca.it
```

```
ssh your_username@login05-ext.leonardo.cineca.it
```

```
ssh your_username@login07-ext.leonardo.cineca.it
```

(For the hackathon, two factor authentication is not used.)

Pixi Package Manager

- Fast, modern, and reproducible package manager
- Can install packages from:
 - PyPI (like pip)
 - conda-forge (like conda)
- <https://pixi.sh/>

Quick intro:

```
curl -fsSL https://pixi.sh/install.sh | bash
pixi init hello-world
cd hello-world
pixi add python # install from conda-forge
pixi add --pypi openai # install from PyPI
pixi run python -c 'print("Hello world!")'
```



Containers on HPC: Singularity / Apptainer

Most HPC systems don't run Docker, but Singularity or Apptainer (which are highly related and behave nearly identically)

<https://docs.sylabs.io/guides/latest/user-guide/>



Convert Docker containers on Leonardo:

```
srun --partition=lrd_all_serial --time 04:00:00 --gres=tmpfs:100G --  
mem=16G --pty singularity pull vllm-openai-v0.21.0-cu129.sif  
docker://docker.io/vllm/vllm-openai:0.21.0-cu129
```

Run something inside a Singularity container:

```
singularity exec --nv --bind $SCRATCH:/scratch container.sif python3
```


Slurm: Job Scheduler

Leonardo job script (1 GPU)

```
#!/bin/bash
#SBATCH --partition=boost_usr_prod
#SBATCH --reservation=s_tra_ncc # Reservation for the hackathon
#SBATCH --nodes=1 # Number of nodes
#SBATCH --ntasks-per-node=1 # Number of `srun` tasks executed per node
#SBATCH --gpus-per-task=1 # Number of GPUs (up to 4 on Leonardo)
#SBATCH --mem=120GB # Fair share on Leonardo: 120GB * gpus-per-task
#SBATCH --cpus-per-task=8 # Fair share on Leonardo: 8 * gpus-per-task
#SBATCH --time=0:30:00 # Time limit in HH:MM:SS, up to 24:00:00

# whatever command should be executed on the compute node.
```

Slurm: Job Scheduler

Leonardo job script (1 GPU)

```
#!/bin/bash
#SBATCH --partition=boost_usr_prod
#SBATCH --reservation=s_tra_ncc # Reservation for the hackathon
#SBATCH --nodes=1 # Number of nodes
#SBATCH --ntasks-per-node=1 # Number of `srun` tasks executed per node
#SBATCH --gpus-per-task=1 # Number of GPUs (up to 4 on Leonardo)
#SBATCH --mem=120GB # Fair share on Leonardo: 120GB * gpus-per-task
#SBATCH --cpus-per-task=8 # Fair share on Leonardo: 8 * gpus-per-task
#SBATCH --time=0:30:00 # Time limit in HH:MM:SS, up to 24:00:00

# Construct run command to execute inside pixi environment:
export RUN_COMMAND="/path/to/pixi run --as-is [--manifest-path pixi_project_path]"

$RUN_COMMAND python3 script.py
```

Slurm: Job Scheduler

Leonardo job script (1 GPU)

```
#!/bin/bash
#SBATCH --partition=boost_usr_prod
#SBATCH --reservation=s_tra_ncc # Reservation for the hackathon
#SBATCH --nodes=1 # Number of nodes
#SBATCH --ntasks-per-node=1 # Number of `srun` tasks executed per node
#SBATCH --gpus-per-task=1 # Number of GPUs (up to 4 on Leonardo)
#SBATCH --mem=120GB # Fair share on Leonardo: 120GB * gpus-per-task
#SBATCH --cpus-per-task=8 # Fair share on Leonardo: 8 * gpus-per-task
#SBATCH --time=0:30:00 # Time limit in HH:MM:SS, up to 24:00:00

# Construct command to run container:
export CONTAINER="singularity exec --nv container.sif"

$CONTAINER python3 script.py
```

Slurm: Job Scheduler

Leonardo job script (2 GPUs)

```
#!/bin/bash
#SBATCH --partition=boost_usr_prod
#SBATCH --reservation=s_tra_ncc # Reservation for the hackathon
#SBATCH --nodes=1 # Number of nodes
#SBATCH --ntasks-per-node=1 # Number of `srun` tasks executed per node
#SBATCH --gpus-per-task=2 # Number of GPUs (up to 4 on Leonardo)
#SBATCH --mem=240GB # Fair share on Leonardo: 120GB * gpus-per-task
#SBATCH --cpus-per-task=16 # Fair share on Leonardo: 8 * gpus-per-task
#SBATCH --time=0:30:00 # Time limit in HH:MM:SS, up to 24:00:00

# Construct command to run container:
export CONTAINER="singularity exec --nv container.sif"

$CONTAINER python3 script.py
```

Slurm: Job Scheduler

Leonardo job script (4 GPUs)

```
#!/bin/bash
#SBATCH --partition=boost_usr_prod
#SBATCH --reservation=s_tra_ncc # Reservation for the hackathon
#SBATCH --nodes=1 # Number of nodes
#SBATCH --ntasks-per-node=1 # Number of `srun` tasks executed per node
#SBATCH --gpus-per-task=4 # Number of GPUs (up to 4 on Leonardo)
#SBATCH --mem=480GB # Fair share on Leonardo: 120GB * gpus-per-task
#SBATCH --cpus-per-task=32 # Fair share on Leonardo: 8 * gpus-per-task
#SBATCH --time=0:30:00 # Time limit in HH:MM:SS, up to 24:00:00

# Construct command to run container:
export CONTAINER="singularity exec --nv container.sif"

$CONTAINER python3 script.py
```

Slurm: Job Scheduler

Leonardo job script (2 nodes with 4 GPUs each)

```
#!/bin/bash
#SBATCH --partition=boost_usr_prod
#SBATCH --reservation=s_tra_ncc # Reservation for the hackathon only enough for
1 node per team
#SBATCH --nodes=2                # Number of nodes
#SBATCH --ntasks-per-node=1      # Number of `srun` tasks executed per node
#SBATCH --gpus-per-task=4        # Number of GPUs (up to 4 on Leonardo)
#SBATCH --mem=480GB              # Fair share on Leonardo: 120GB * gpus-per-task
#SBATCH --cpus-per-task=32       # Fair share on Leonardo: 8 * gpus-per-task
#SBATCH --time=0:30:00           # Time limit in HH:MM:SS, up to 24:00:00

# Construct command to run container:
export CONTAINER="singularity exec --nv container.sif"

srun $CONTAINER python3 script.py
```

Slurm: Job Scheduler

Useful Slurm commands

Submit a job:

\$ sbatch job.sh

Check submitted jobs:

\$ squeue --me

Look at the output from a job:

\$ cat slurm-<job_id>.out

or follow the output as the job runs:

\$ tail -c +0 -f slurm-<job_id>.out

Cancel job:

\$ scancel <job_id>

Get a shell at a node while a job is running:

\$ srun --overlap --pty --jobid=<job_id> bash

Leonardo

File storage:

Shared storage between all login and compute nodes.

- \$HOME: 50 GB limit
- \$SCRATCH: Higher limit. Use this for larger files during the hackathon. (Files are deleted after 40 days.)
- \$PUBLIC: 50 GB limit. Can be used to share files between Leonardo users.
- \$FAST and \$WORK: Do not use during the hackathon.

Login node CPU time limit:

Process on login nodes have a 10 minute CPU time limit.

Use this for longer processes:

```
srun --partition=lrd_all_serial --time 04:00:00 --gres=tmpfs:100G --mem=16G  
--pty bash
```



Leonardo internet access

- Use login nodes for large file downloads.
- Compute nodes do not have internet access.
- As a workaround, set the following environment variables in your Slurm script:

```
export HTTP_PROXY=http://proxyuser:5dd1d2bd00@10.99.0.1:38425
export HTTPS_PROXY=http://proxyuser:5dd1d2bd00@10.99.0.1:38425
export http_proxy=http://proxyuser:5dd1d2bd00@10.99.0.1:38425
export https_proxy=http://proxyuser:5dd1d2bd00@10.99.0.1:38425
```

The proxy will restart every once in a while (due to the 10 min CPU time limit).
TCP connections will drop shortly.

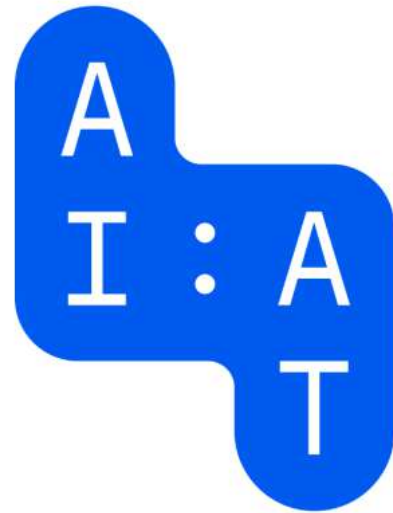
Please only use the proxy for low-bandwidth traffic. Always download large files from the login nodes.

Additional Material

Our HPC Onboarding Kit contains additional information:

- Chapter 5: First steps on LEONARDO
- Chapter 6: Software

<https://ai-at.eu/hpc-onboarding/>



Supercomputer Onboarding Kit

An AI:AT Tutorial



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LET THE HACK BEGIN!